

Submission & Reviewer Matching

Weekly update from Qiang

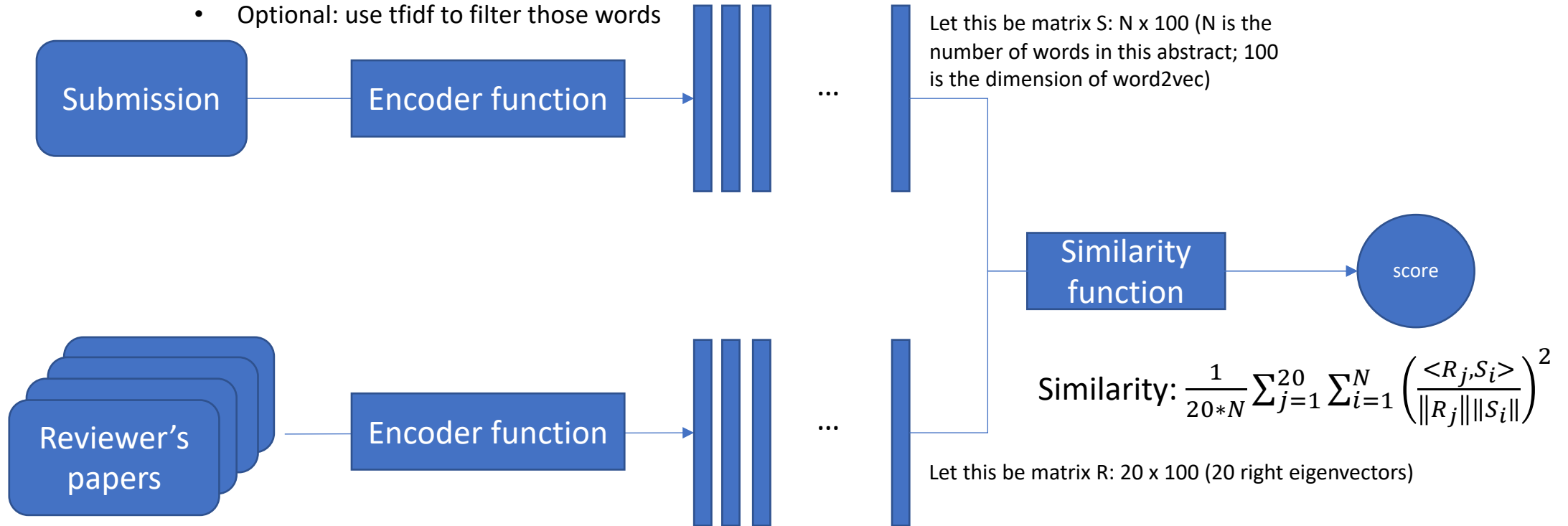
07/16/2019

Update

1. Initial try of TFIDF
2. Issues of our evaluation metric

What we have now

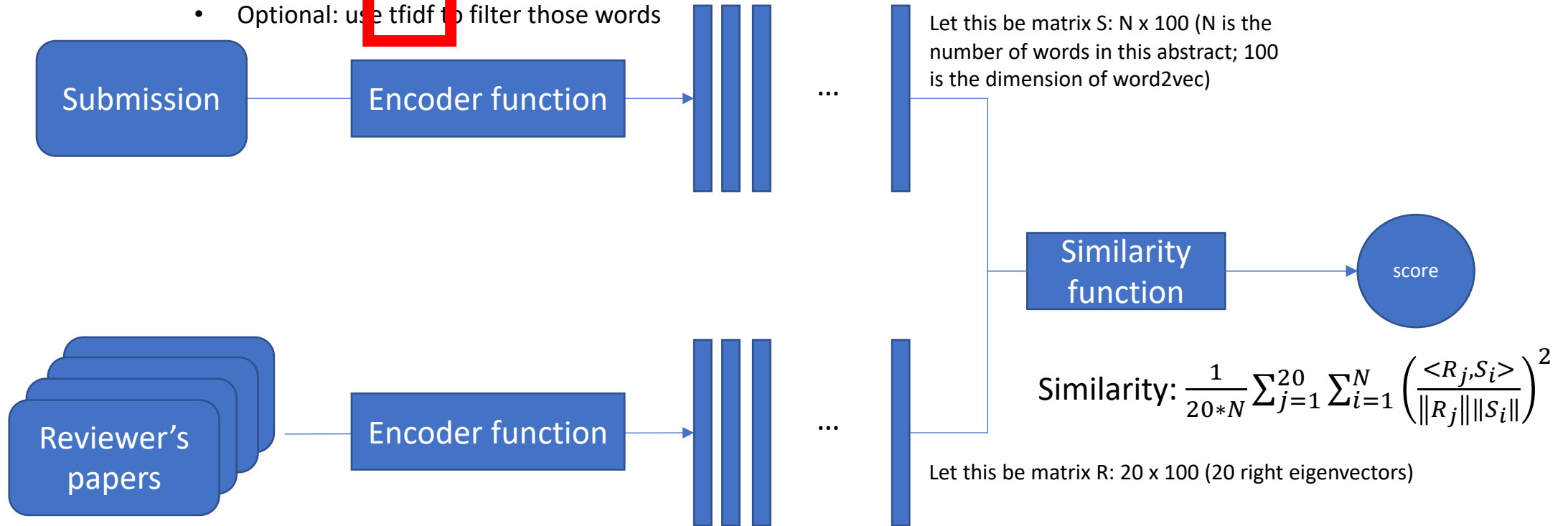
- Pre-train word2vec on all the papers in our corpus
- Stack all the word vectors from the abstract of the submission
- Optional: use tfidf to filter those words



- Pre-define a weight vector for papers in specific years (putting more weight on more recent papers)
- Stack all the word vectors from the abstract of the papers of one particular review
- Perform SVD and obtain the first 20 right eigenvectors
- Optional: use tfidf to filter those words

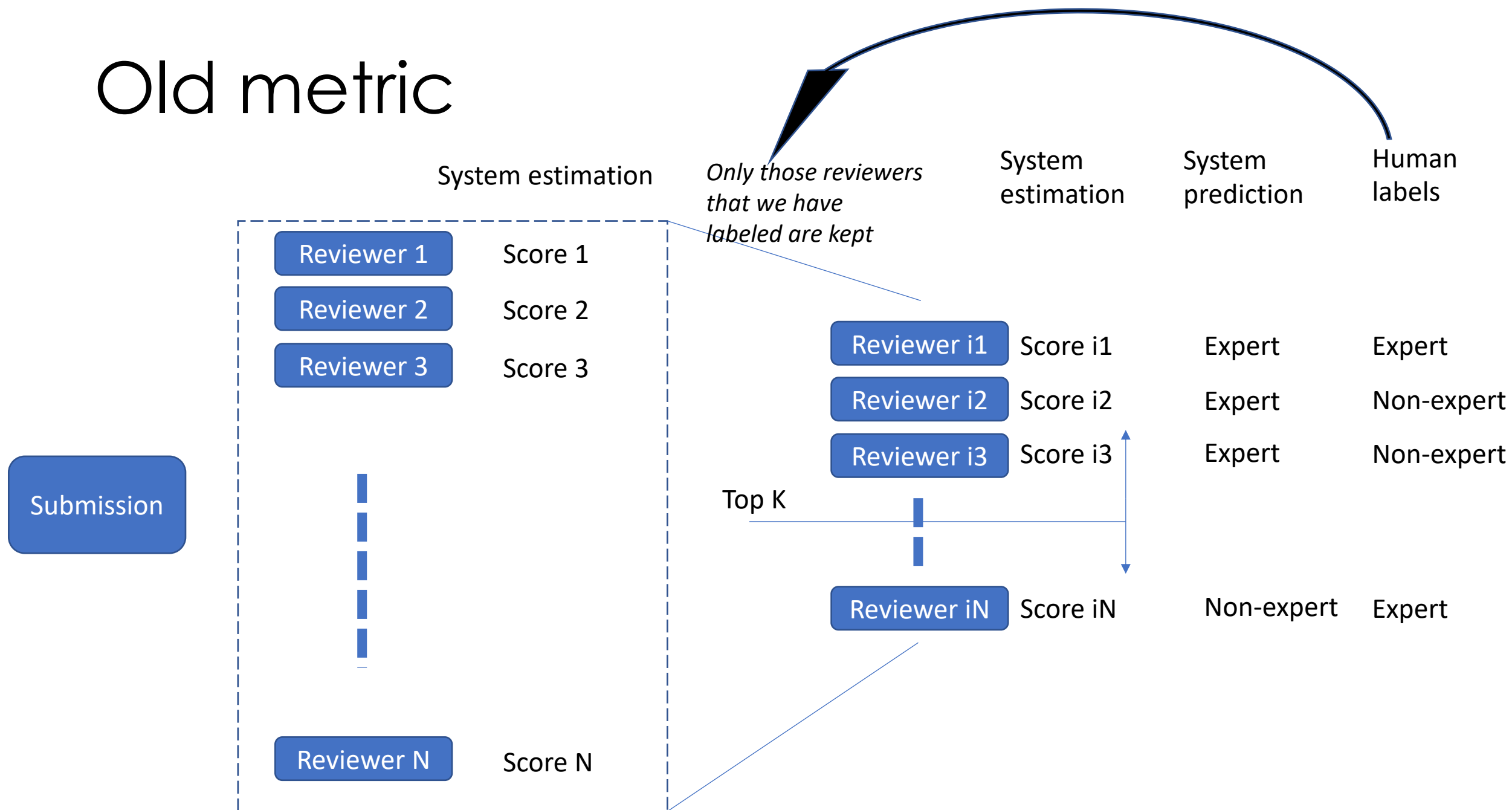
What we have now + TFIDF

- Pre-train word2vec on all the papers in our corpus
- Stack all the word vectors from the abstract of the submission
- Optional: use tfidf to filter those words

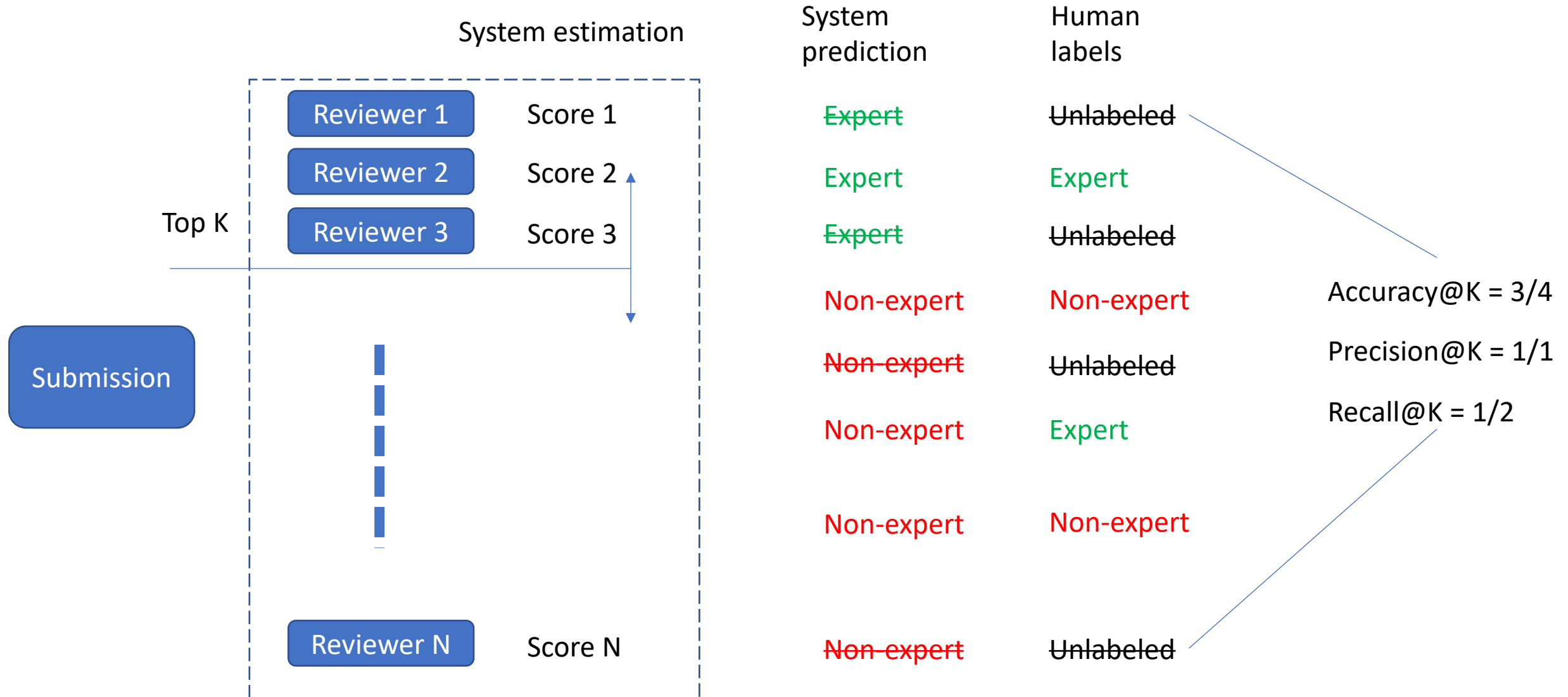


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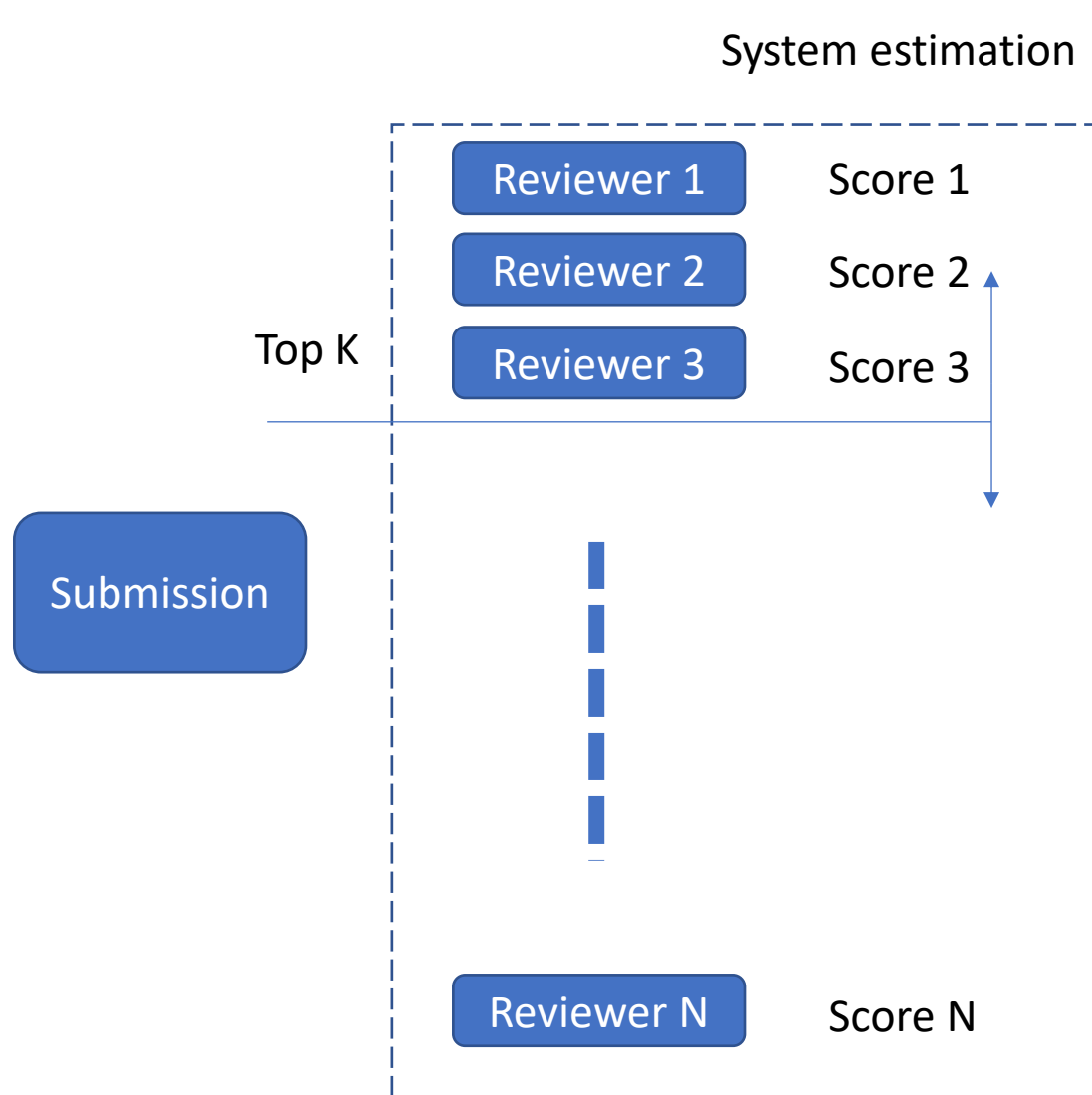
Old metric



Metrics at top K



Mean rank



System prediction

~~Expert~~

Expert

~~Expert~~

Non-expert

~~Non-expert~~

Non-expert

Non-expert

~~Non-expert~~

Human labels

Unlabeled

Expert

Unlabeled

Non-expert

Unlabeled

Expert

Non-expert

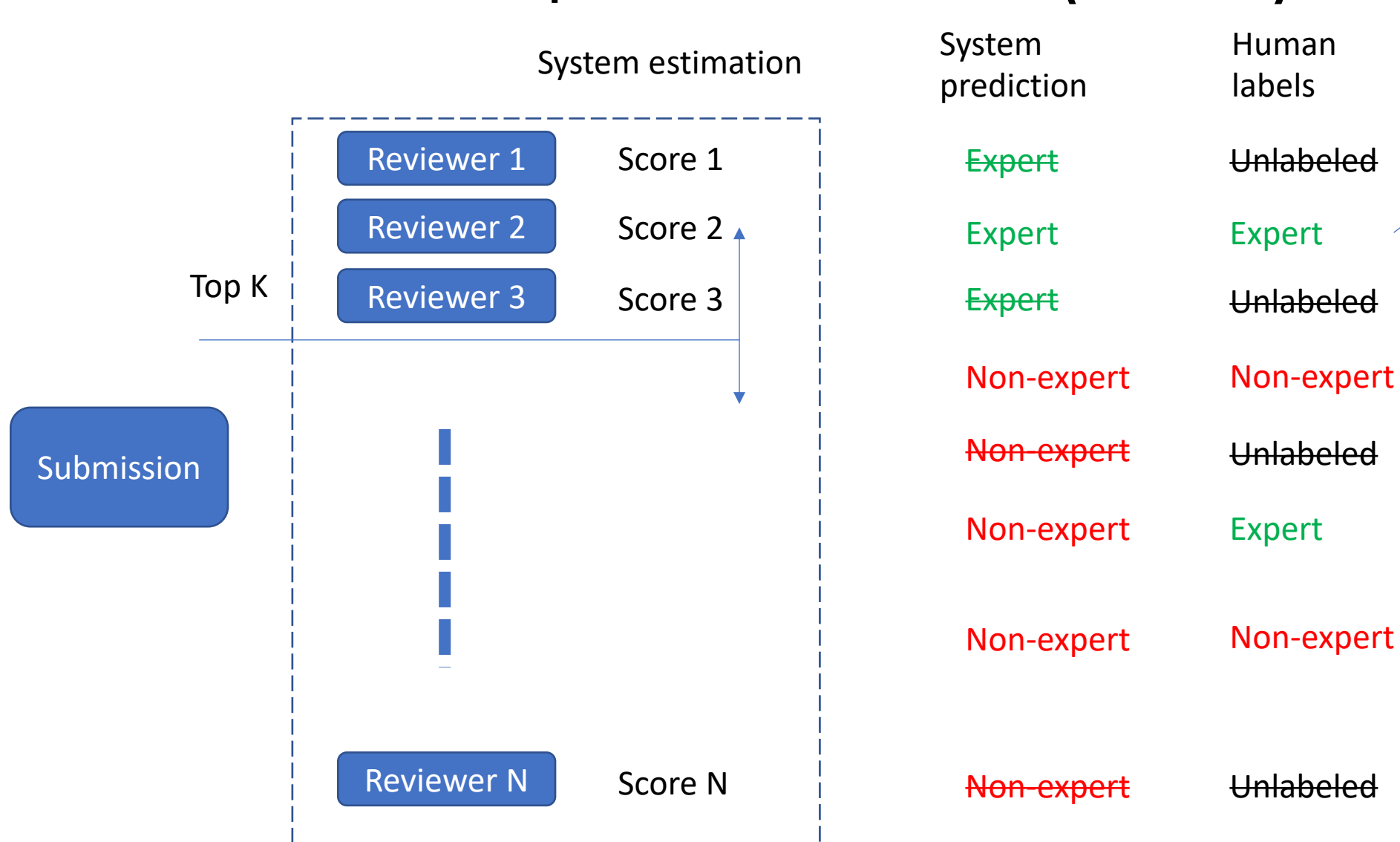
Unlabeled

$$MR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} rank_i$$

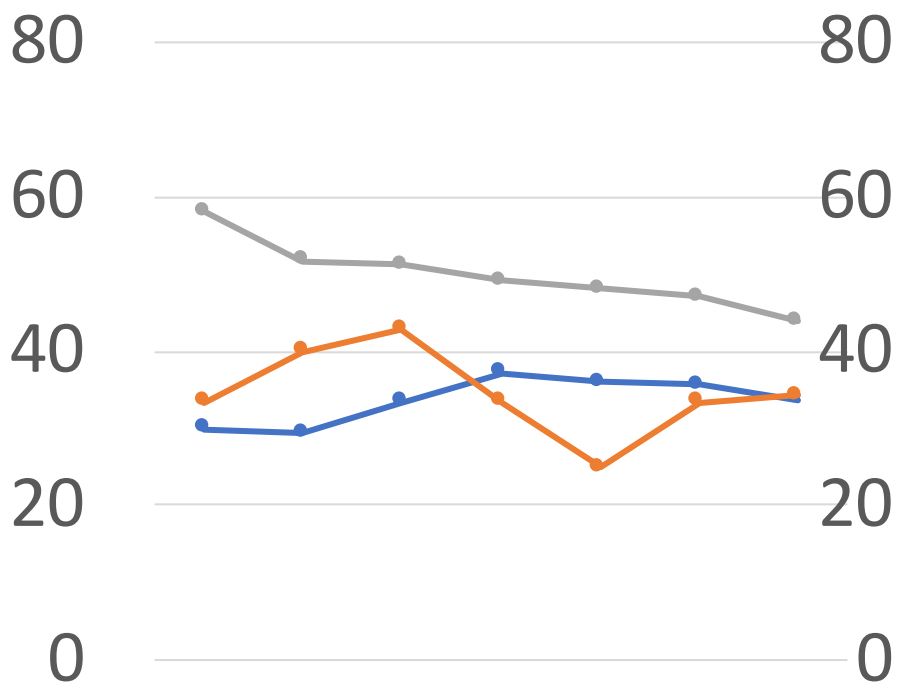
Penalizes too much when an expert is ranked very low.

Mean reciprocal rank (MRR)

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}$$



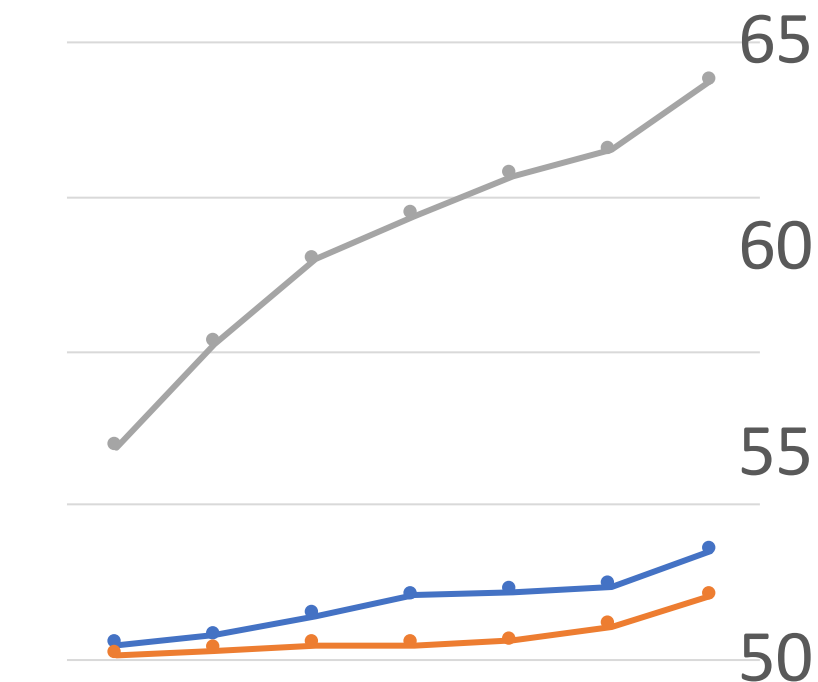
Precision



kappa= 5 10 15 20 25 30 50

—random
—current
—current+tfidf

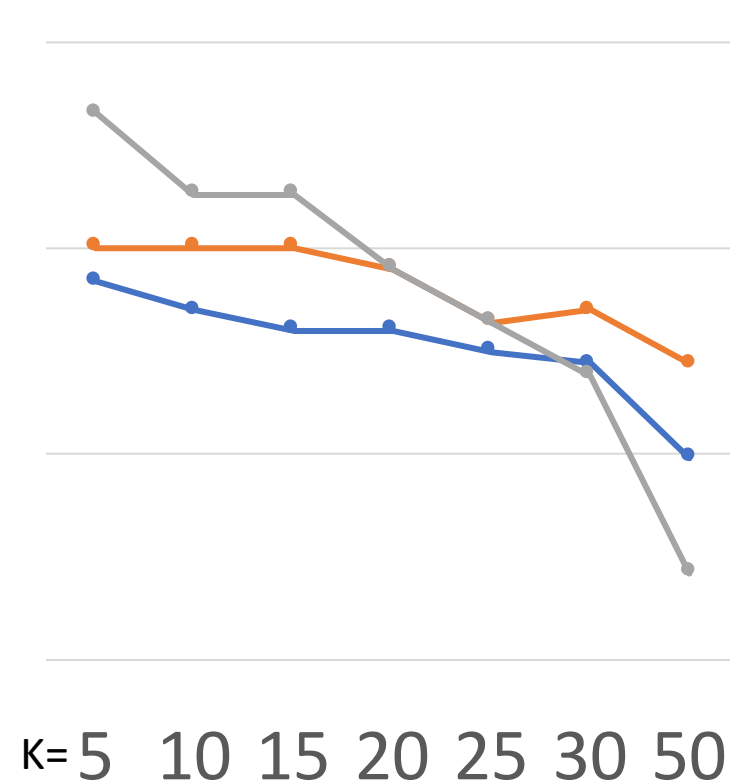
Recall



kappa= 5 10 15 20 25 30 50

—random
—current
—current+tfidf

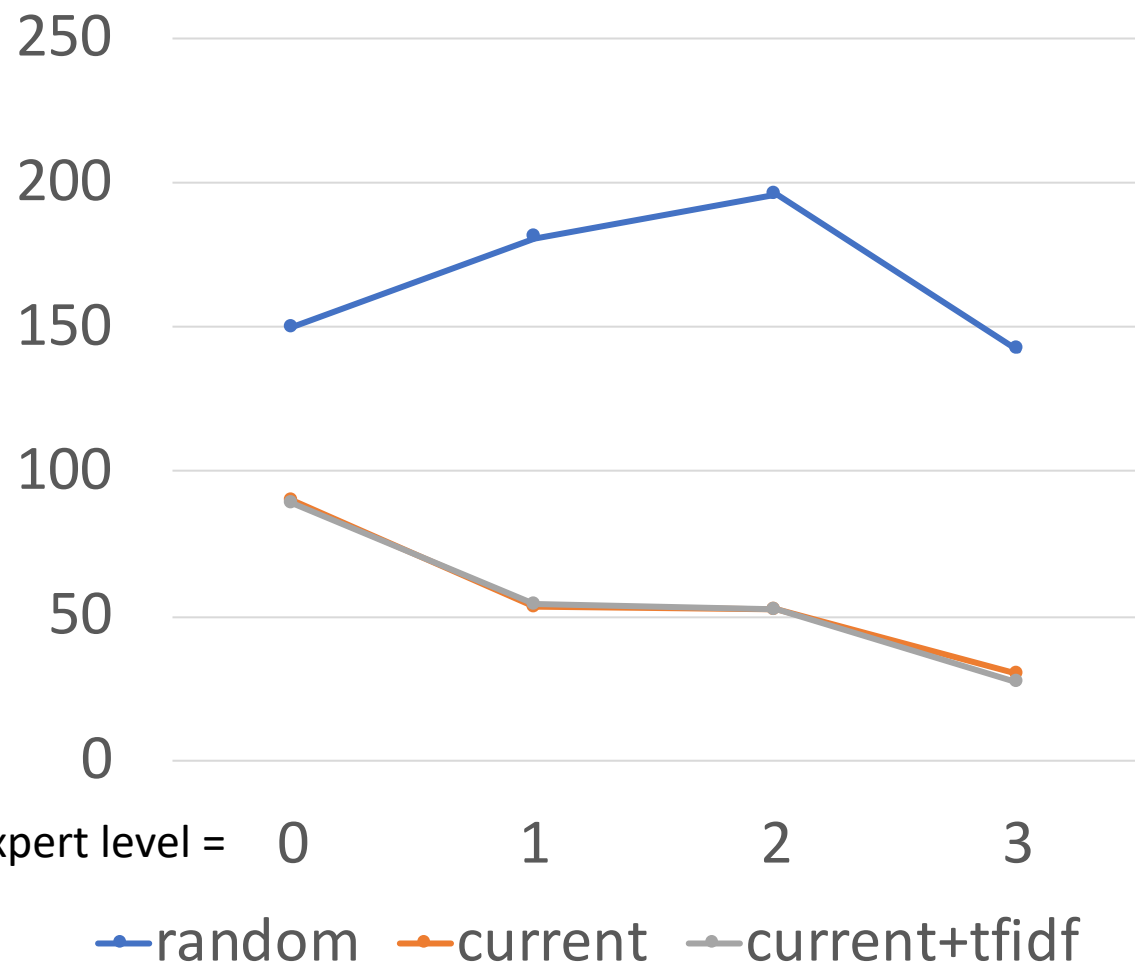
Accuracy



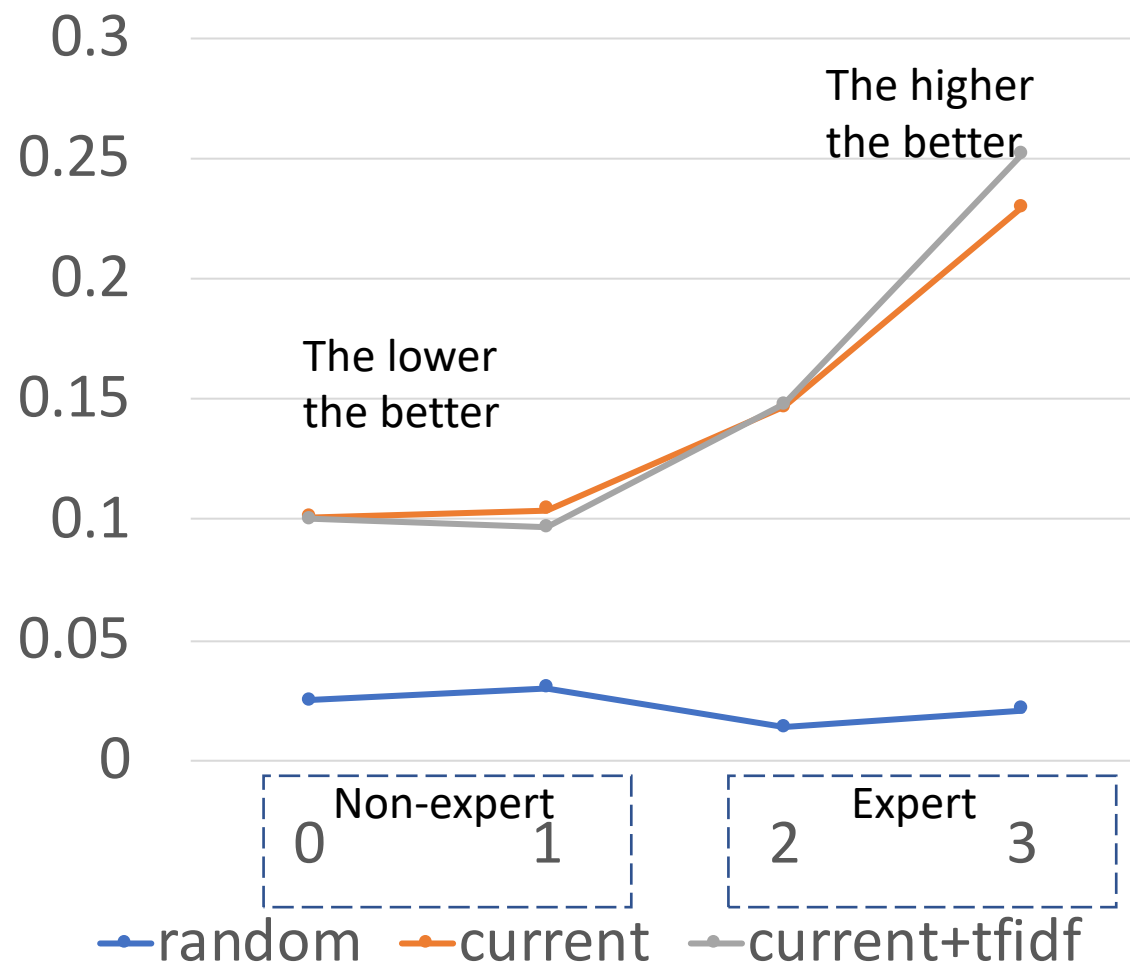
kappa= 5 10 15 20 25 30 50

—random
—current
—current+tfidf

Mean rank



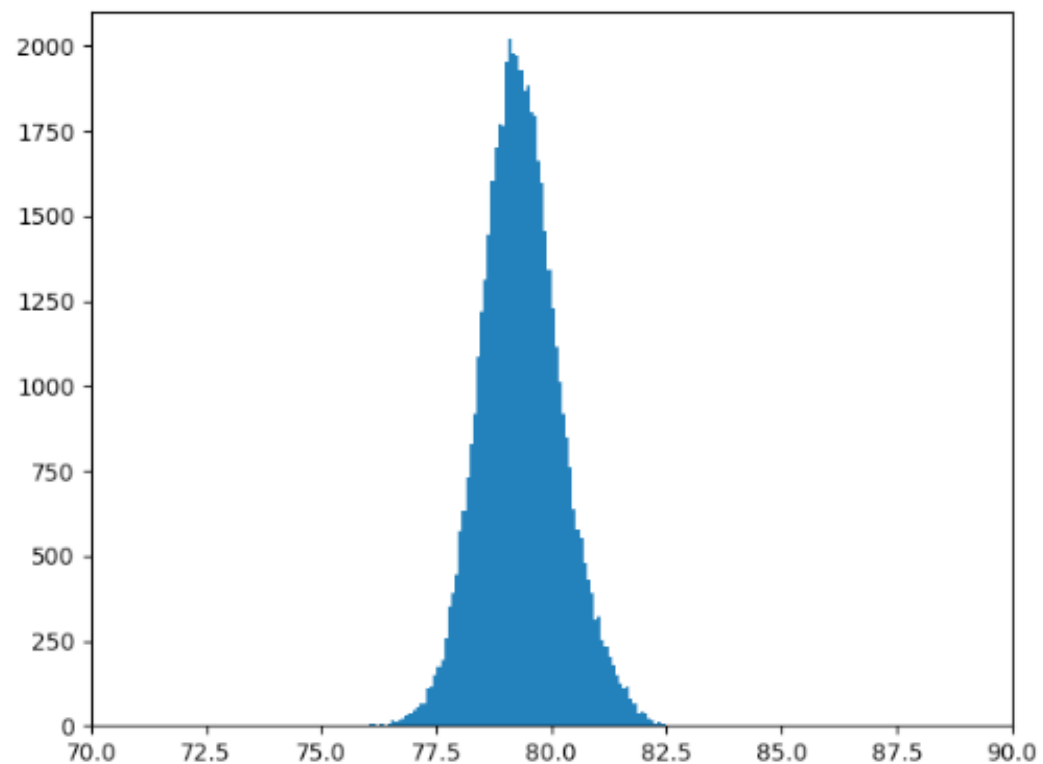
MRR



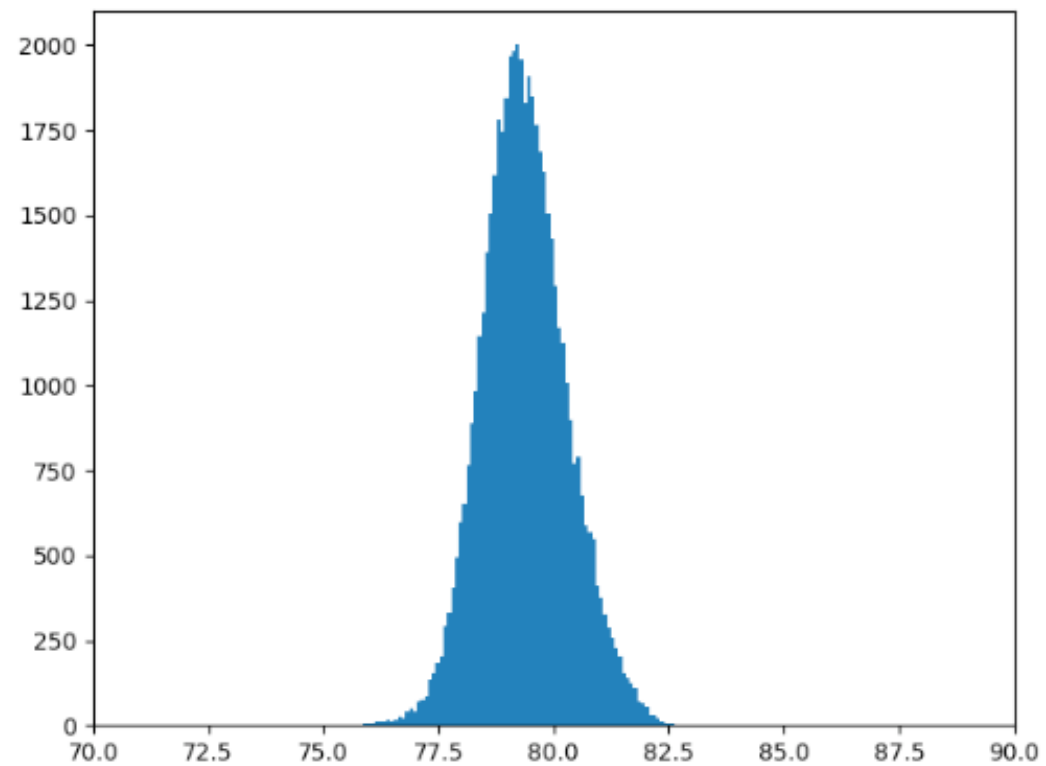
Old metric

- Random: 0.440000000000000001
 - Current: 0.540000000000000001
 - Current+TFIDF: 0.539999999999999999
- While it is clear that Current+TFIDF is better than Current using the new metrics, our old metric wasn't able to show this.

Current



Current+TFIDF



The concentration issue wasn't solved by TFIDF.